

Confidence

Intervals

Agenda

1) Null Hypotheses Revisited

2) CI's

↳ RBI

Two-way ANOVA Table

$$Y_i = \mu + \alpha_{j(i)} + \beta_{k(i)} + \sigma_{j(i), k(i)} + \epsilon_i, \\ \epsilon_i \sim N(0, \sigma^2)$$

Source	df	SS	MS	F	H ₀
Factor A	J-1	SS _A	MS _A	MS _A /MS _{resid}	
Factor B	K-1	SS _B	MS _B	MS _B /MS _{resid}	
Interaction A:B	(J-1)(K-1)	SS _{AB}	MS _{AB}	MS _{AB} /MS _{resid}	
Residual	n - JK	SS _{res}	MS _{resid}		

RBI

[Slides]

Confidence Intervals

def (LB, UB) A random interval that contains the parameter of interest w/ Probability $\geq (1-\alpha)$ over whatever random process generated the data.

Q - What is our random data generative process?

id $y_i(0)$ $y_i(1)$

1

15

2

2

30

2

3

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40

4

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